

Architecture-Dependent Decoherence Suppression in Passive Quantum Networks: Irreducible Channel Mixing, Squared Rate Gaps, and the Price in Dwell Time

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Abstract

Can passive reservoir filtering suppress decoherence more strongly when its channel mixing is irreducible, even after passband responses and rational complexity are fixed? We give an explicit affirmative separation, together with the resource that pays for it. For every integer $S \geq 5$ we construct a causal three-signal/three-loss-port inner transfer whose signal block G_S is a rational Schur function with numerator degree at most $S + 4$ and common denominator degree S . It obeys $3(S - 1)$ exact delayed tangential calibrations in a pass arc; every three calibration signatures span $\mathbb{C}^3$. On two stop arcs,

$$\sup \|G_S\|_{\text{op}} \leq \varepsilon_S := \left(\frac{\pi S}{6} + \frac{1}{2} \right) e^{-9S/20}.$$

In contrast, every transfer of the same bidegree that has a constant nontrivial reducing channel and obeys the same calibrations has stopband norm at least one. The proof is a scalar root-count obstruction exposed by the full-spark signatures. The lossless completion uses only three scalar Blaschke all-pass elements, balanced interferometers and paraunitary delays, so physical passivity is verified directly rather than inferred from an unconstrained interpolation. For a bath spectral density $j_- I \preceq J \preceq j_+ I$, the filtered Kossakowski matrix therefore has norm at most $j_+ \varepsilon_S^2$, whereas every reducing comparator retains a directional rate at least j_- . This is a squared exponential separation at the dissipative-rate level. It is not free: if a finite Blaschke product of degree S yields signal leakage at most $\varepsilon < 1$ on an arc of length L , its peak group delay is at least $LS/(16 \arcsin \varepsilon)$. Our family has exponentially large delay. Thus under a peak-delay budget D , the rate-improvement factor is at most $\csc^2(LS/(16D))$, asymptotically quadratic in D/S . Irreducible passive mixing can therefore move a coherence-maintenance boundary, but it moves the required resource into long-lived passive memory rather than erasing it. Exact arithmetic, dense-grid unitarity audits and all figures are reproduced by the linked public source package.

1 Question, result, and non-claims

Quantum filter functions translate control protocols into frequency-resolved noise response [1, 2], and passive linear quantum systems admit network and realization theories that preserve the canonical commutation relations [3, 4, 5]. Tangential interpolation can itself be made physical-realizability- and passivity-preserving [6]. Blaschke–Potapov factors have also been used to identify trapped modes in delayed passive quantum networks [7]. None of those ingredients is claimed as new.

The narrower question here is architectural. Suppose that a multichannel filter must match many directional responses exactly, has a fixed rational bidegree, and remains passive. Is a device whose channels split after one frequency-independent rotation necessarily as capable as one whose channel geometry changes with frequency? The answer is no. We prove an exponential signal-level

gap between an explicit irreducible signal block and every constant-channel reducing comparator. Passing that gap through a uniformly nondegenerate bath spectral density squares the suppression exponent. We then prove a delay lower bound that prevents the architecture theorem from being misread as a free thermodynamic advantage.

This result develops the resource-boundary viewpoint of [8] in a direction orthogonal to spatial buffer thickness. The protected object is a frequency-resolved coupling channel. The filter can move the rate boundary by changing its channel architecture, but poles near the unit circle store the resource as dwell time. The paper makes no claim to derive the Born rule, select outcomes, solve an interacting many-body buffer, or show that a small thermodynamic lower bound is achievable.

The contributions are:

1. an explicit six-port causal inner completion, with no appeal to a black-box synthesis theorem;
2. a fixed-bidegree irreducible/reducible separation under exact full-spark tangential calibrations;
3. a squared separation for filtered Kossakowski matrices;
4. a general delay-price theorem for the scalar all-pass mechanism;
5. an exact no-go showing why the earlier real-symmetric quadratic certificate cannot be made globally passive by scalar rescaling; and
6. a reproducible ledger linking every numerical statement to code [9].

2 Passive transfer model and comparison class

We use the unit-disc convention. A matrix transfer $S(z)$ is *inner* when it is analytic for $|z| < 1$ and unitary almost everywhere on $\mathbb{T} = \{z : |z| = 1\}$. A principal block $G(z)$ of an inner transfer is Schur: $\|G(z)\|_{\text{op}} \leq 1$ in the disc and on its regular boundary points. In a passive quantum-optical interpretation, the omitted ports carry vacuum loss; unitarity of the full annihilation-operator transfer preserves the canonical commutation relations. Delays and stable all-pass factors are included, so the model is a discrete-time or sampled traveling-field network, not an instantaneous scattering matrix.

Let

$$\mathcal{I}_p = [-\pi/6, \pi/6], \quad \mathcal{I}_s = [-\pi, -5\pi/6] \cup [5\pi/6, \pi].$$

Given boundary nodes $z_j = e^{i\theta_j}$ and unit signatures $v_j \in \mathbb{C}^3$, the prescribed response is the two-sample delay

$$Q(z_j)v_j = z_j^2 v_j. \tag{1}$$

This common delay makes the calibration compatible with causality and with a lossless pass response.

Definition 2.1 (Rational bidegree class). For $S \geq 1$, let $\mathcal{R}_S$ be the set of 3×3 rational transfers $Q = A/d$, regular on $\mathbb{T}$, whose scalar common denominator satisfies $\deg d \leq S$ and whose numerator entries satisfy $\deg A_{ab} \leq S + 4$. The class allows cancellations; using upper bounds rather than exact degrees only strengthens the comparator.

Definition 2.2 (Constant-channel reducibility). A transfer Q is *reducing* if a nonzero proper subspace $E \subset \mathbb{C}^3$, independent of z , reduces $Q(z)$ at every regular point: both E and $E^\perp$ are invariant. In dimension three, either E or its orthogonal complement contains a reducing line $\mathbb{C}u$. Thus for a unit u there is a scalar rational branch q with

$$Q(z)u = q(z)u, \quad Q(z)^*u = \overline{q(z)}u \quad (z \in \mathbb{T}).$$

This comparator is strictly larger than simultaneously diagonalizable or commuting channels: the complementary 2×2 block may be fully noncommutative and frequency dependent. The comparison therefore isolates a constant channel split, not mere pairwise commutativity.

3 Why the earlier zero-phase witness cannot simply be made passive

The earlier matrix-polynomial construction [10] used four real-symmetric quadratic tangential constraints. It is tempting to rescale that witness until its norm is at most one. The following exact calculation shows that the idea fails before any quantum interpretation is attempted.

Let

$$x = (1/2, 1, 3/2, 7/2), \quad v = ((1, 2), (0, 1), (-1, 2), (-1, 1)),$$

and impose $P(x_j)v_j = v_j$ on a real-symmetric quadratic P . Define

$$R(x) = R_0 + xR_1 + x^2R_2$$

with

$$R_0 = \begin{pmatrix} 5/6 & -1/3 \\ -1/3 & 5/6 \end{pmatrix}, \quad R_1 = \begin{pmatrix} -2/3 & -1/3 \\ -1/3 & -11/6 \end{pmatrix}, \quad R_2 = \begin{pmatrix} 2/3 & 2/3 \\ 2/3 & 1 \end{pmatrix}.$$

Proposition 3.1 (Exact Hermitian passivity no-go). *Every feasible real-symmetric quadratic is $P_\tau = I + \tau R$. If $\|P_\tau(x)\|_{\text{op}} \leq 1$ for every real x , then $\tau = 0$.*

Proof. In the symmetric-entry coordinates $(a_k, b_k, c_k)_{k=0}^2$, the eight scalar calibration equations have rational rank eight. Direct substitution shows that $R(x_j)v_j = 0$, so the affine solution space is exactly $I + \tau R$. Now $R(0)$ is positive definite because its leading entry is $5/6$ and its determinant is $7/12$. Hence $\tau > 0$ gives an eigenvalue of $P_\tau(0)$ strictly above one. At $x = 5/2$,

$$R(5/2) = \begin{pmatrix} 10/3 & 3 \\ 3 & 5/2 \end{pmatrix}, \quad \det R(5/2) = -2/3,$$

so $R(5/2)$ has a negative eigenvalue; $\tau < 0$ again makes an eigenvalue of $P_\tau(5/2)$ exceed one. Only $\tau = 0$ remains. $\square$

The obstruction motivates the boundary-phase construction below. Its calibration is delayed rather than zero phase, its coefficients are complex causal transfer coefficients rather than Hermitian polynomials on the real line, and passivity is built into a lossless completion.

4 Explicit Blaschke–interferometer construction

Fix $\alpha = 9/20$ and $S \geq 5$. Put

$$r_S = 1 - e^{-\alpha S}, \quad b_r(z) = \frac{z - r}{1 - rz}, \quad B_S(z) = \eta_S b_{r_S}(z)^S, \quad (2)$$

where $\eta_S = 1$ for odd S and $\eta_S = -1$ for even S . Thus $B_S(-1) = -1$. Set $\delta_S = 1 - r_S$ and

$$\phi_g = (g - 1)\delta_S, \quad p_g(z) = \frac{1 + e^{i\phi_g} B_S(z)}{2}, \quad g = 0, 1, 2. \quad (3)$$

Let $F_{ab} = 3^{-1/2}\omega^{ab}$ with $\omega = e^{2\pi i/3}$, and define paraunitary mixers

$$U(z) = F \operatorname{diag}(1, z, z^2) F^*, \quad U_{\text{rev}}(z) = F \operatorname{diag}(z^2, z, 1) F^*. \quad (4)$$

On $\mathbb{T}$, $U_{\text{rev}}(z) = z^2 U(z)^*$. The signal transfer is

$$G_S(z) = U(z) \operatorname{diag}(p_0(z), p_1(z), p_2(z)) U_{\text{rev}}(z). \quad (5)$$

4.1 Direct six-port lossless completion

For $B_g(z) = e^{i\phi_g} B_S(z)$ define the balanced Mach–Zehnder block

$$W_g(z) = \frac{1}{2} \begin{pmatrix} 1 + B_g(z) & 1 - B_g(z) \\ 1 - B_g(z) & 1 + B_g(z) \end{pmatrix} = H \operatorname{diag}(1, B_g(z)) H, \quad H = 2^{-1/2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}. \quad (6)$$

Each W_g is inner and has signal entry p_g . Let W place the three W_g blocks on the signal/loss pairs $(g, g+3)$. Then

$$S_S(z) = \operatorname{diag}(U(z), I_3) W(z) \operatorname{diag}(U_{\text{rev}}(z), I_3) \quad (7)$$

is inner, and its upper-left 3×3 block is G_S .

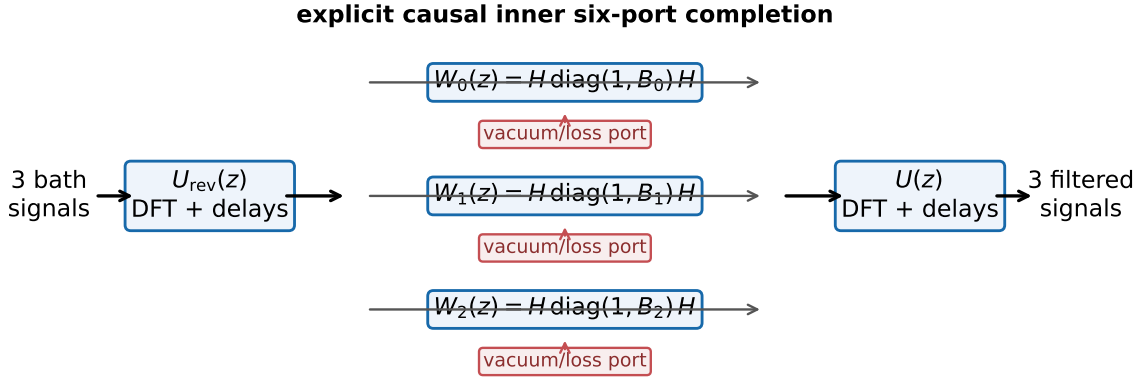


Figure 1: Component-level realization. The signal passes through a reverse paraunitary mixer, three parallel lossless Mach–Zehnder/all-pass blocks and a forward mixer. The three auxiliary inputs are vacuum/loss ports. Every displayed block is inner, so their cascade is inner.

Proposition 4.1 (Physical passivity by construction). *S_S is a causal rational inner six-port transfer and G_S is a rational Schur transfer in $\mathcal{R}_S$.*

Proof. b_r is analytic in the unit disc and unimodular on $\mathbb{T}$. Hence every B_g , W_g , W , U , and U_{rev} is inner; products and block diagonals preserve innerness. The signal-block contraction follows from the unitarity of the boundary completion. The common denominator is $(1 - r_S z)^S$. The scalar p_g numerators have degree at most S , while the two mixers add at most four, so the numerator entries of G_S have degree at most $S + 4$. $\square$

The factorization is also a component prescription: three repeated scalar all-pass elements, three balanced interferometers, a three-mode Fourier network, and integer delays. General passive-network realization results [5, 11] provide a broader context, but are not needed for the boundary unitarity identity used here.

4.2 Pass nodes and full-spark signatures

At a point where $B_g(z) = 1$, let

$$v_{g,z} = U(z) e_g. \quad (8)$$

Then $p_g(z) = 1$ and the identity $U_{\text{rev}} = z^2 U^*$ gives

$$G_S(z) v_{g,z} = z^2 v_{g,z}. \quad (9)$$

Lemma 4.2 (Node count). *For each $g \in \{0, 1, 2\}$, the open pass arc contains at least $S - 1$ distinct solutions of $B_g(z) = 1$.*

Proof. On the unit circle write $b_r(e^{i\theta}) = e^{i\beta_r(\theta)}$ with a continuous unwrapped phase on $(-\pi, \pi)$ and

$$\beta'_r(\theta) = \frac{1 - r^2}{1 - 2r \cos \theta + r^2} > 0.$$

For $r = r_S$, put $\beta_0 = \beta_r(\pi/6)$. The identity

$$\beta_0 = 2 \arctan\left((2e^{9S/20} - 1) \tan \frac{\pi}{12}\right)$$

and $\cot x < 1/x$ show that $\beta_0 > \pi(1 - 1/S)$ for every $S \geq 5$: the sufficient inequality $(2e^{9S/20} - 1) \tan(\pi/12) > 2S/\pi$ holds at $S = 5$ and its left-minus-right derivative is positive thereafter. More directly, the inverse formula

$$\theta = 2 \arctan\left(\frac{1 - r}{1 + r} \tan \frac{\beta}{2}\right)$$

maps the interval $(-\beta_0, \beta_0)$ into the pass arc. The allowed interval for the integer winding number in $S\beta + \arg \eta_S + \phi_g = 0 \pmod{2\pi}$ has length $S\beta_0/\pi > S - 1$. Every open real interval of length greater than $S - 1$ contains at least $S - 1$ integers, independently of the phase offset. Hence at least $S - 1$ roots are interior. $\square$

Select any $S - 1$ nodes from each group and call the resulting set $\mathcal{Z}_S$; its size is $M = 3(S - 1)$. These signatures have a stronger property than spanning collectively.

Lemma 4.3 (Full spark). *Every three distinct signatures in $\{v_{g,z} : (g,z) \in \mathcal{Z}_S\}$ are linearly independent.*

Proof. Multiplication by F^* does not change independence, and

$$F^*v_{g,z} = 3^{-1/2}(1, t, t^2)^\top, \quad t = \omega^{-g}z.$$

The three pass-node groups are rotated by $0, -2\pi/3, -4\pi/3$; their arcs are disjoint, so distinct signatures give distinct t . The determinant of any three columns is the nonzero Vandermonde product $3^{-3/2}(t_2 - t_1)(t_3 - t_1)(t_3 - t_2)$. $\square$

4.3 Stopband leakage

Lemma 4.4 (Analytic stop envelope). *On $\mathcal{I}_s$,*

$$\|G_S(e^{i\theta})\|_{\text{op}} \leq \varepsilon_S := \left(\frac{\pi S}{6} + \frac{1}{2}\right) e^{-9S/20}. \quad (10)$$

Proof. The mixers are unitary on $\mathbb{T}$, so the norm is $\max_g |p_g|$. The all-pass phase derivative is S times the Poisson kernel. On either stop arc, $\cos \theta \leq -\sqrt{3}/2$, whence

$$\frac{1 - r^2}{1 - 2r \cos \theta + r^2} \leq 1 - r^2.$$

Starting from $B_S(-1) = -1$, its phase deviates from π by at most $S(1 - r^2)\pi/6$. Including $|\phi_g| \leq 1 - r$ and using $|(1 + e^{i(\pi+x)})/2| \leq |x|/2$ gives

$$|p_g| \leq \frac{\pi S}{12}(1 - r^2) + \frac{1}{2}(1 - r) \leq \left(\frac{\pi S}{6} + \frac{1}{2}\right)(1 - r).$$

Substitute $1 - r = e^{-\alpha S}$. $\square$

5 The architecture-separation theorem

We can now state the central result.

Theorem 5.1 (Passive irreducible/reducing separation). *Fix $S \geq 5$ and $\alpha = 9/20$. There are $M = 3(S - 1)$ pass nodes in $\mathcal{I}_p$ and full-spark signatures such that:*

1. *the signal block $G_S \in \mathcal{R}_S$ has the explicit six-port inner completion (7), obeys all calibrations (1), and has stop leakage at most ε_S ;*
2. *every reducing $Q \in \mathcal{R}_S$ obeying the same calibrations satisfies*

$$\|Q(z)\|_{\text{op}} \geq 1 \quad (z \in \mathcal{I}_s); \quad (11)$$

3. *if such a comparator is Schur, equality holds in (11) throughout the stopband;*
4. *whenever $\varepsilon_S < 1$, G_S is irreducible.*

Thus the signal-level architecture ratio is at least $1/\varepsilon_S$ and is exponential up to the displayed polynomial prefactor.

Proof. The constructive statements follow from Theorems 4.1 to 4.4. It remains to prove the comparator obstruction. Let $\mathbb{C}u$ be a reducing line of Q and let q be its scalar branch. Left-projecting each calibration with u^* gives

$$(q(z_j) - z_j^2)\langle u, v_j \rangle = 0. \quad (12)$$

Full spark implies that at most two signatures lie in the plane $u^\perp$; otherwise three would be dependent. Consequently, $q - z^2$ vanishes at at least

$$M - 2 = 3S - 5$$

distinct boundary points. After placing q over the common denominator d , the numerator of $q - z^2$ has degree at most $S + 4$. Since $3S - 5 > S + 4$ for $S \geq 5$, the numerator is identically zero. Therefore $q(z) = z^2$ wherever regular, and $\|Q(z)\|_{\text{op}} \geq \|Q(z)u\| = 1$. A Schur comparator supplies the reverse inequality. Finally, if G_S were reducing, the same result would force its stop norm to be at least one, contradicting $\varepsilon_S < 1$. $\square$

Remark 5.2 (What is held fixed). Both sides have three signal channels, identical delayed directional calibrations and the same rational bidegree ceiling. The irreducible side is also required to possess an explicit passive completion. The lower bound is proved for a larger comparison class that need not be passive; imposing passivity only turns its lower bound into equality.

Remark 5.3 (Exact rather than robust separation). The theorem uses exact calibrations and exact constant reducing subspaces. Small calibration errors can be treated by polynomial conditioning estimates, but the present full-spark margin decreases rapidly with S ; we do not claim a uniform constant-error theorem. This limitation is operationally relevant and is revisited in Section 10.

6 Transfer to bath spectra and decoherence rates

Let $J(\omega)$ be the 3×3 bath spectral-density matrix in a protected frequency band. Assume the uniform nondegeneracy condition

$$j_- I \preceq J(\omega) \preceq j_+ I, \quad 0 < j_- \leq j_+ < \infty. \quad (13)$$

For a linear passive prefilter T , the output noise spectrum is

$$\Gamma_T(\omega) = T(e^{i\theta(\omega)})J(\omega)T(e^{i\theta(\omega)})^*. \quad (14)$$

In a weak-coupling Davies construction, Γ_T is the Kossakowski matrix sampled at the relevant Bohr frequency [12]; in stationary Gaussian pure dephasing it is directly the quadratic rate matrix. The following statement is linear algebra and therefore does not depend on a particular master-equation normalization.

Corollary 6.1 (Squared Kossakowski gap). *On the stopband, the constructed signal block obeys*

$$\|\Gamma_{G_S}(\omega)\|_{\text{op}} \leq j_+ \varepsilon_S^2. \quad (15)$$

For every reducing calibrated comparator $Q \in \mathcal{R}_S$, there is a fixed unit direction u such that

$$u^* \Gamma_Q(\omega) u \geq j_- \quad (16)$$

throughout the stopband. Hence the worst retained directional rate divided by the constructed worst-case rate is at least $j_-/(j_+ \varepsilon_S^2)$.

Proof. Equation (15) follows from $\|TJT^*\| \leq \|T\|^2 \|J\|$. For the comparator, the proof of Theorem 5.1 gives $Q^*u = \bar{z}^2 u$ on $\mathbb{T}$. Thus

$$u^* Q J Q^* u = (Q^*u)^* J (Q^*u) \geq j_- \|Q^*u\|^2 = j_-.$$

□

For a transition whose three coupling matrix elements form a vector c , its Davies or pure-dephasing quadratic susceptibility is proportional to $c^* \Gamma_T c$. Consequently Theorem 6.1 gives an exact rate separation for any model in which the relevant transition can probe the retained direction u . It also gives an architecture-independent upper bound for all unit coupling vectors under G_S .

In pointer-basis pure dephasing, the coherence lifetime is inversely proportional to the corresponding quadratic rate. The construction therefore permits a lifetime gain of order ε_S^{-2} relative to the retained comparator direction, subject to the Markov, stationarity and spectral-bounds assumptions above. Applying a maintenance inequality of the form $P_{\text{extra}} \geq k_B T \dot{C}_{\text{loss}}$, as carefully delimited in [8], transfers the rate comparison to the *input of the lower bound*. A smaller lower bound does not prove an achievable low-power maintenance protocol, and we do not use it that way.

7 The hidden resource: a delay-price theorem

Exponential suppression in Theorem 5.1 is obtained by taking r_S exponentially close to one. This creates long-lived internal modes. The next result shows that this is not an accident of our parameterization.

Let

$$B(z) = \xi \prod_{k=1}^S \frac{z - a_k}{1 - \bar{a}_k z}, \quad |\xi| = 1, \quad |a_k| < 1,$$

and $p = (1 + B)/2$. On the unit circle let $\tau_B(\theta)$ denote the unwrapped phase derivative, or group delay in sample units:

$$\tau_B(\theta) = \sum_{k=1}^S \frac{1 - |a_k|^2}{|e^{i\theta} - a_k|^2}. \quad (17)$$

Theorem 7.1 (Stop suppression forces dwell time). *Let $I \subset \mathbb{T}$ be a connected arc of angular length $L > 0$. If $\sup_I |p| \leq \varepsilon < 1$, then*

$$\sup_{\theta \in [-\pi, \pi]} \tau_B(\theta) \geq \frac{LS}{16 \arcsin \varepsilon}. \quad (18)$$

Proof. The condition $|1 + B| \leq 2\varepsilon$ confines the phase of B to an arc of half-width $2 \arcsin \varepsilon$ around π . Because I is connected and $\varepsilon < 1$, a continuous unwrapped phase cannot change branches inside I ; its total variation is at most $4 \arcsin \varepsilon$. On the other hand, each Poisson kernel in (17) obeys the uniform lower bound

$$\frac{1 - |a_k|^2}{|e^{i\theta} - a_k|^2} \geq \frac{1 - |a_k|}{4}.$$

Integration over I therefore yields

$$\frac{L}{4} \sum_{k=1}^S (1 - |a_k|) \leq 4 \arcsin \varepsilon.$$

For some k_* , $1 - |a_{k_*}| \leq 16 \arcsin(\varepsilon)/(LS)$. At $\theta = \arg a_{k_*}$, its Poisson kernel is $(1 + |a_{k_*}|)/(1 - |a_{k_*}|)$, which alone is at least the right-hand side of (18). $\square$

Corollary 7.2 (Delay-budget frontier). *Under the hypotheses of Theorem 7.1, let $D = \sup_{\theta} \tau_B(\theta)$. Then*

$$\varepsilon \geq \sin\left(\frac{LS}{16D}\right). \quad (19)$$

Consequently, if $\delta = \varepsilon^2$ is the corresponding rate envelope, the rate-improvement factor satisfies

$$\frac{1}{\delta} \leq \csc^2\left(\frac{LS}{16D}\right) \sim \left(\frac{16D}{LS}\right)^2 \quad (D/S \rightarrow \infty). \quad (20)$$

In particular, exponential rate improvement requires exponential peak delay when L is fixed and S grows subexponentially relative to the target.

Proof. Rearrange (18) to obtain $\arcsin \varepsilon \geq LS/(16D)$ and use monotonicity of sine on $[0, \pi/2]$. Squaring and inverting gives (20); the asymptotic form follows from $\sin x \sim x$. $\square$

For the repeated real zero in (2), the exact peak is

$$\tau_{B_S}^{\max} = S \frac{1 + r_S}{1 - r_S} = S(2e^{\alpha_S} - 1). \quad (21)$$

Thus both the general lower bound and the construction are exponential when the leakage is exponential, though the simple repeated-zero design is not claimed constant-optimal. In physical language, high-Q passive memory replaces active pumping. That is a real architectural benefit in settings where storage time is cheap and drive power is expensive, but not a removal of resource cost. Theorem 7.2 makes the trade explicit: the squared spectral-rate advantage can grow at most quadratically with available peak delay, up to the order and arc-length factors shown.

8 Numerical and exact verification

The proofs above are analytic. Computation audits implementation identities, finite-precision behavior and scale. The public replay performs:

1. rational Gaussian elimination for Theorem 3.1;
2. closed-form pass-node generation and all $\binom{M}{3}$ Vandermonde minor checks;
3. dense global contractivity and stopband sweeps;
4. direct 6×6 unitarity tests on 1001 boundary frequencies;
5. the rational root-count inequality; and

6. the squared-rate and group-delay identities.

No optimizer is used to establish the theorem.

For the audit we fix $\alpha = 0.45$. Table 1 reports the observed stop norm ρ_S , its square, the reciprocal rate factor, the exact peak delay, and the maximum six-port unitarity residual. The smallest full-spark minor remains positive but becomes poorly conditioned by $S = 19$; this is why the numerical study stops there rather than presenting meaningless larger orders.

S	M	ρ_S	ρ_S^2	$1/\rho_S^2$	$\tau_{\max}$
5	15	$1.269 \cdot 10^{-1}$	$1.610 \cdot 10^{-2}$	$6.21 \cdot 10^1$	$8.99 \cdot 10^1$
7	21	$6.245 \cdot 10^{-2}$	$3.900 \cdot 10^{-3}$	$2.56 \cdot 10^2$	$3.20 \cdot 10^2$
9	27	$2.990 \cdot 10^{-2}$	$8.939 \cdot 10^{-4}$	$1.12 \cdot 10^3$	$1.02 \cdot 10^3$
11	33	$1.402 \cdot 10^{-2}$	$1.965 \cdot 10^{-4}$	$5.09 \cdot 10^3$	$3.09 \cdot 10^3$
15	45	$2.940 \cdot 10^{-3}$	$8.643 \cdot 10^{-6}$	$1.16 \cdot 10^5$	$2.56 \cdot 10^4$
19	57	$5.895 \cdot 10^{-4}$	$3.475 \cdot 10^{-7}$	$2.88 \cdot 10^6$	$1.96 \cdot 10^5$

Table 1: Dense-grid audit at $\alpha = 0.45$. All maximum calibration residuals are below 6.3×10^{-12} and all six-port unitarity errors below 6.0×10^{-14} . The theorem uses only $3(S - 1)$ nodes; the replay audits all $3S$ interior nodes found for these orders.

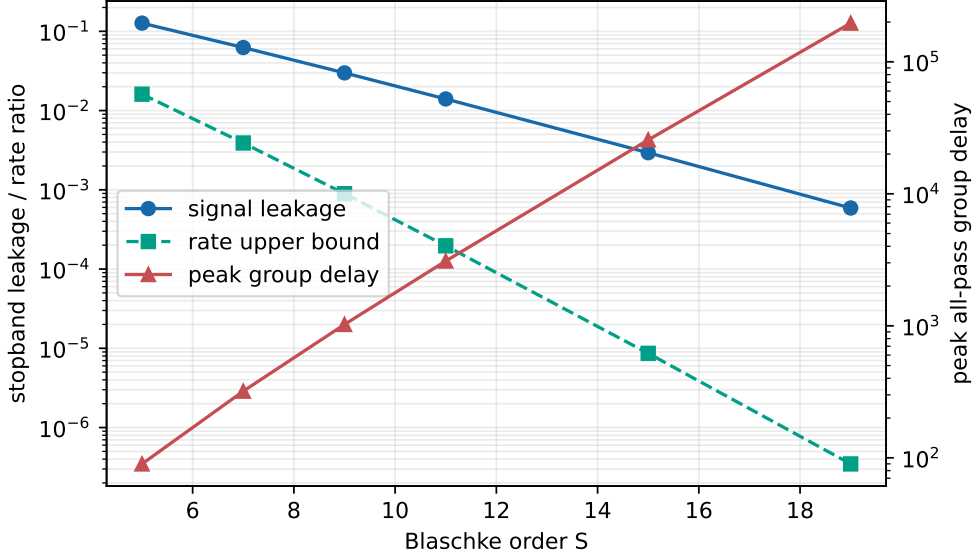


Figure 2: Signal leakage and its squared rate envelope fall exponentially, while peak all-pass group delay rises exponentially. Lines connect audited orders and are not statistical fits.

The observed rate factor exceeds 2.8×10^6 at $S = 19$, but that number should be read beside a peak delay near 2.0×10^5 samples and a minimum Vandermonde minor near 1.3×10^{-14} . The result is a theorem about architecture and asymptotic possibility, not a claim that the largest audited device is robustly fabricable.

8.1 Passive-preserving tolerance stress test

To distinguish algebraic existence from component tolerance, we run a frozen Monte Carlo stress test at $S = 11$. Each of 2000 trials independently perturbs the logarithm of each pole gap $1 - r_g$ by a Gaussian standard deviation of 1%, each arm phase by 1% of the nominal pole gap, and each balanced interferometer angle by 10^{-3} radians. The seed is 20260810. All perturbations preserve $|B_g| = 1$ and use orthogonal beam splitters, so every trial remains a passive lossless six-port network; only its nominal calibration is lost.

The median stop norm is 0.01421, compared with the nominal 0.01402; its 95th percentile is 0.01474 and maximum is 0.01611. In contrast, the maximum residual over the *nominal* pass nodes has median 0.0700 and 95th percentile 0.1293. Thus stop suppression is locally stable in this component model, while exact high-order calibration is sensitive to pole placement. Retuning the pass frequencies restores exact roots for each lossless sample, but a fixed-frequency application must budget this error. The stress test supports the resource/conditioning caveat; it is not evidence for a robust comparator separation.

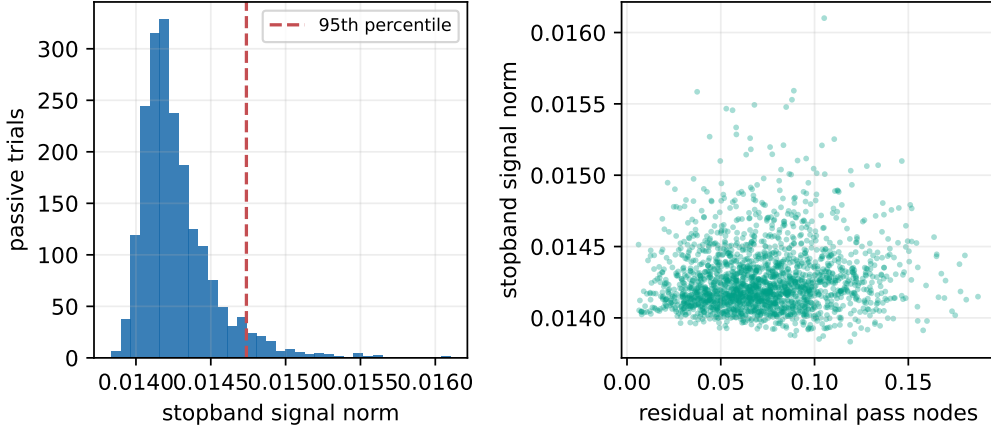


Figure 3: Passive-preserving tolerance audit at $S = 11$ (2000 frozen trials). Stop leakage changes mildly, whereas residuals at the nominal calibration nodes expose the high-Q phase sensitivity.

9 Relation to prior work and novelty boundary

Filter-function theory already supplies frequency-domain descriptions of controlled open-system error [1, 2], and experimental noise filtering is established [13]. Passive quantum feedback networks, lossless bounded-real transfers and cascade realizations are likewise established [3, 4, 5, 11]. Coherent passive reservoir engineering has, in particular, been formulated as a pole-placement route to decoherence-free subsystems [14]. Most directly, Techakesari and Nurdin preserve tangential responses and passivity in model reduction [6]; therefore neither “tangential quantum interpolation” nor “physical interpolation” is a novelty claim here. Tabak and Mabuchi connect Blaschke–Potapov factorization to trapped modes in delayed passive quantum networks [7]; therefore the use of Blaschke factors and the interpretation of near-boundary poles as trapped memory are also prior art.

The new mathematical claim is the conjunction proved in Theorems 5.1, 6.1, 7.1 and 7.2: under one explicit collection of full-spark delayed calibrations and one fixed rational bidegree, a passive irreducible signal block has exponentially small stop leakage while every constant-channel reducing transfer retains unit leakage; a uniformly nondegenerate bath squares this gap at the rate-matrix level; and strong scalar all-pass suppression forces a quantitative peak-delay cost. The earlier matrix-polynomial paper [10] supplied the classical root-count architecture mechanism. Here passivity, causal boundary phase, an explicit quantum-compatible loss completion, the Kossakowski transfer and the delay price change both the theorem and its interpretation. Delay–bandwidth and storage–bandwidth tradeoffs are classical themes; the paper claims neither theme in general. The candidate novelty is the explicit inequality used together with the fixed-bidegree architecture and bath-rate separations.

10 Limitations and falsification routes

Limitation 10.1 (Exact calibration). The separation can be fragile because the minimum full-spark minor decreases with order. A fabrication-relevant theorem should replace exact roots by a quantified approximate root count and propagate phase, pole and calibration uncertainties through the comparator lower bound.

Limitation 10.2 (Linear stationary bath interface). Equation (14) assumes a linear filter before a stationary Gaussian spectral density. General non-Gaussian or nonlinear reservoirs need not transform by congruence, and strong-coupling or non-Markovian dynamics need not be summarized by a Davies rate matrix.

Limitation 10.3 (No many-body mass-gap theorem). The construction is a rational passive network, not an interacting quantum field theory. It neither proves a Yang–Mills mass gap nor establishes the interacting-buffer conjecture in [8]. Its relevance is methodological: it exhibits an exactly solvable architecture-dependent rate boundary and the memory resource behind it.

Limitation 10.4 (Restricted comparator). The no-go covers constant reducing subspaces at the stated bidegree. A reducible device with higher rational degree, frequency-dependent channel coordinates, feedback outside the modeled ports, active gain, or relaxed calibrations lies outside the theorem. These are possible escape routes, not contradictions.

The most direct experimental falsification is a passive three-channel device that satisfies the same calibrated responses and bidegree but has a verified constant reducing line and stop norm below one. Such a device would contradict the algebraic root count and should be detectable from the public calibration ledger. A failure of the squared rate law under the stated linear-Gaussian assumptions would instead falsify the bath-transfer model.

11 Conclusion

Irreducible channel mixing is not merely a coordinate choice when calibration directions vary with frequency. An explicit passive six-port family exploits that fact to produce exponentially small signal leakage at a rational complexity where every constant-channel reducing comparator retains a unit branch. The bath spectral density squares the separation in dissipative rates. The same construction also exposes its own bill: exponentially long group delay, hence high-Q passive memory. Operationally, the Heisenberg cut can be displaced by architecture, but the resource boundary reappears as dwell time and conditioning. That simultaneous positive result and resource accounting is the central claim.

Code and data availability

Source, exact-arithmetic checks, dense numerical audits, the JSON ledger and the scripts that regenerate Figs. 1 to 3 are available in the public repository [9]:

Public source and verification branch

The principal commands are

```
python research/quantum_reservoir_filter.py
python research/quantum_reservoir_robustness.py
python verification/verify_passive_quantum_filter.py
```

The release record pins the PDF and result-ledger hashes. No private data, remote service or stochastic seed is required.

A A sharper view of the root-count obstruction

Suppose a reducing comparator has a subspace of dimension two rather than a declared reducing line. Its orthogonal complement is a reducing line because both the subspace and its complement are invariant by definition. Therefore the proof of Theorem 5.1 loses at most two calibrations, regardless of which nontrivial reducing dimension is initially supplied.

The degree $S + 4$ is also conservative. If the scalar branch numerator has lower degree due to cancellation, the root-count contradiction becomes stronger. Boundary roots are legitimate because the denominator is assumed regular on $\mathbb{T}$. The proof does not require the comparator to be stable, causal, passive or quantum realizable; those restrictions only shrink its class.

B Bath congruence in a pure-dephasing model

For completeness, consider coupling operators $A_a = A_a^*$ and bath fields $B_a = B_a^*$ with interaction $H_I = \lambda \sum_{a=1}^3 A_a \otimes B_a$. At a fixed secular frequency, collect the bath correlations in $J_{ab}(\omega)$. A linear prefilter with annihilation-amplitude transfer T maps the field vector to TB (with the loss ports initially in vacuum), so the signal contribution to its correlation matrix is TJT^* . For a transition $m \leftrightarrow n$, let $c_a = \langle m|A_a|n \rangle$. The associated quadratic dissipative coefficient is, up to the common convention-dependent prefactor, c^*TJT^*c . This gives (14). Vacuum loss ports can add a baseline when the physical observable includes their fluctuations; the theorem concerns suppression of the specified incoming bath contribution, and any such baseline must be added separately.

For commuting pointer observables, the same calculation applies at zero Bohr frequency to contrasts between pointer states. If the contrast vector is c , the off-diagonal density-matrix element decays as $\exp[-\kappa(c)t]$ with $\kappa(c)$ proportional to $c^*\Gamma_T(0)c$. This is the exact pure-dephasing setting in which the resource-boundary interpretation is cleanest.

C Reproducibility ledger

The JSON record stores, for each S :

- r_S , node count, numerator degree and root-count margin;
- minimum determinant magnitude over all signature triples;
- maximum calibration residual;
- global and stopband signal norms;
- the squared rate envelope and reciprocal factor;
- exact peak group delay; and
- maximum six-port unitarity residual.

The analytic acceptance gates are deliberately looser than floating-point roundoff: calibration residual below 10^{-9} , unitarity residual below 10^{-10} , global norm below $1 + 2 \times 10^{-12}$, positive full-spark minors, and stop norm below (10). A failed gate exits nonzero.

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